

Walmart Store Sales Forecasting to Minimize Loss

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11th April 2026

Abstract

Objective: This study investigates optimal inventory decisions under stochastic demand. **Method:** A convex optimization model based on the Newsvendor framework is formulated using Walmart sales data. **Results:** The optimal order quantity is derived via the critical fractile, showing sensitivity to cost asymmetry and demand variability. **Conclusion:** Stochastic modeling significantly improves inventory efficiency, though limitations arise from distributional assumptions.

1 Problem

In retail operations, managing inventory under uncertain demand is a classic challenge that directly impacts a company's bottom line. Overstocking capitalizes cash and incurs holding costs, while understocking leads to stockouts, lost sales, and diminished customer loyalty. The context of this research problem centers on Walmart, a retail giant where external factors such as markdowns and holiday events cause significant volatility in weekly sales. Understanding and mathematically modeling this volatility is crucial for optimal asset allocation.

This optimization project is motivated by the intersection of quantitative modeling and operational efficiency. By treating demand not as a static figure but as a probability distribution, we can minimize expected operational costs.

Research Question (RQ): How can we formulate and solve a convex optimization problem to determine the optimal weekly inventory level for a specific Walmart department that minimizes the expected total cost of overstocking and understocking, given a stochastic demand distribution?

2 Plan

To answer the research question, we will measure the historical weekly sales data of Walmart stores and utilize it to estimate the parameters of our demand distribution.

Data Collection and Sampling

The data utilized is secondary data sourced from the Kaggle platform, specifically the “Walmart Recruiting – Store Sales Forecasting” dataset. The data was originally collected internally by Walmart through their point-of-sale systems and made publicly available for research and predictive modeling.

The sampling frame consists of 45 Walmart stores located in various regions, encompassing weekly records from February 2010 to November 2012. The data represents a census of these specific 45 stores over the defined time horizon rather than a probabilistic sample of all global Walmart stores. The dataset is large, containing over 400,000 rows across 99 departments.

Weblink to Data: <https://www.kaggle.com/c/walmart-recruiting-store-sales-forecasting/data>

3 Data

The raw dataset comprises several variables distributed across three tables: store information (Store, Type, Size), external features (Temperature, Fuel_Price, CPI, Unemployment, Markdown1–5), and the target variable (Weekly_Sales).

Data Cleaning and Preparation

1. **Handling Missing Values:** The Markdown variables (Markdown1–5) contained a substantial number of missing values, as Walmart only began recording these promotional features from November 2011 onwards. Instead of discarding these observations, the missing entries were imputed with zeros, which reasonably represent weeks without promotional markdown activity. Missing values in macroeconomic indicators such as CPI and Unemployment were handled using forward-fill and backward-fill techniques to preserve temporal continuity and avoid artificial discontinuities in the time series.
2. **Treatment of Anomalies:** Observations with negative Weekly_Sales were removed from the dataset. These values correspond to product returns rather than actual consumer demand. Since the Newsvendor framework assumes non-negative demand, retaining such values would violate model assumptions and distort the estimated demand distribution.
3. **Filtering and Variable Isolation:** To construct a tractable and well-defined optimization problem, the dataset was filtered to isolate a single time series corresponding to Store 1 and Department 1. This allows for focused modeling while preserving the essential stochastic characteristics of demand.
4. **Time-Series Structuring:** The data was arranged in chronological order based on weekly timestamps. This step ensures consistency for subsequent statistical analysis and facilitates the identification of temporal patterns such as seasonality and holiday-driven demand spikes.
5. **Data Validation and Consistency Checks:** After preprocessing, the dataset was validated through summary statistics and visual inspection. This included verifying the absence of missing values, confirming the non-negativity of sales, and ensuring consistency across merged datasets.

Descriptive Statistics

To better understand the behavior of demand, we examine key summary statistics of Weekly Sales for Store 1, Department 1.

- Mean (μ): 22513.32
- Standard Deviation (σ): 9854.35
- Minimum: 14537.37
- Maximum: 57592.12
- Skewness: 2.01
- Kurtosis: 3.36

The mean and standard deviation provide empirical estimates for the parameters of the assumed demand distribution. The large gap between the maximum and mean values indicates the presence of significant demand spikes, particularly during holiday periods.

The positive skewness suggests that the distribution is right-skewed, implying that extreme high-demand realizations occur more frequently than extreme low-demand ones. Additionally, the kurtosis value greater than 3 indicates the presence of heavy tails, meaning that extreme events are more likely than predicted under a perfect Normal distribution.

These observations highlight that while the Normal distribution assumption provides analytical tractability, real-world demand exhibits deviations such as skewness and heavy tails. This further justifies the use of a stochastic modeling framework to account for demand uncertainty.

4 Analysis

Mathematical Formulation

Cost Function Representation:

The cost incurred for an order quantity Q and realized demand D is defined as:

$$C(Q, D) = c_h(Q - D)^+ + c_s(D - Q)^+$$

where

$$(Q - D)^+ = \max(Q - D, 0), \quad (D - Q)^+ = \max(D - Q, 0)$$

Here,

- c_h is the holding cost per unit (cost minus salvage value)
- c_s is the shortage cost per unit (selling price minus cost)

Standardization:

Let

$$z = \frac{Q - \mu}{\sigma}$$

Then the expected cost formulation can be expressed in terms of the standard normal cumulative distribution function $\Phi(z)$ and probability density function $\phi(z)$, enabling efficient numerical computation. The core of this analysis relies on formulating a stochastic continuous-review inventory model (the Newsvendor model). Let our decision variable be Q , the quantity of inventory to stock.

1. **Random Variable:** Demand is not deterministic; it is a continuous random variable denoted as D . Based on historical data for Store 1, Department 1, we assume D follows a Normal distribution with mean μ and standard deviation σ . Its probability density function is $f(x)$.

2. **Cost Parameters:**

- c_h : Holding cost per unit of unsold inventory (overage)
- c_s : Shortage/penalty cost per unit of unmet demand (underage)

3. **Expectation-Type Objective Function:**

The objective is to minimize the expected total cost:

$$\min_Q \mathbb{E}[C(Q)] = c_h \int_0^Q (Q - x)f(x) dx + c_s \int_Q^\infty (x - Q)f(x) dx$$

Subject to:

$$Q \geq 0$$

4. **Convex Optimization:**

Taking the first derivative:

$$\frac{d\mathbb{E}[C(Q)]}{dQ} = c_h F(Q) - c_s (1 - F(Q))$$

Taking the second derivative:

$$\frac{d^2\mathbb{E}[C(Q)]}{dQ^2} = (c_h + c_s)f(Q)$$

Since $c_h, c_s > 0$ and $f(Q) \geq 0$, the function is strictly convex, guaranteeing a global minimum.

Optimal Order Quantity (Newsvendor Solution)

To determine the optimal inventory level, we set the first derivative of the expected cost function equal to zero:

$$\frac{d\mathbb{E}[C(Q)]}{dQ} = c_h F(Q) - c_s(1 - F(Q)) = 0$$

Solving for $F(Q)$, we obtain:

$$F(Q^*) = \frac{c_s}{c_s + c_h}$$

This ratio is known as the *critical fractile* and represents the optimal service level that balances the trade-off between overstocking and understocking.

Assuming that demand follows a Normal distribution with mean μ and standard deviation σ , the optimal order quantity can be expressed as:

$$Q^* = \mu + z^* \sigma$$

where z^* satisfies:

$$\Phi(z^*) = \frac{c_s}{c_s + c_h}$$

and $\Phi(\cdot)$ denotes the cumulative distribution function of the standard normal distribution.

This formulation provides a closed-form solution for the optimal inventory level based on estimated demand parameters and cost structure.

Economic Interpretation

The optimal solution depends critically on the relative magnitude of shortage and holding costs, as captured by the critical fractile:

$$F(Q^*) = \frac{c_s}{c_s + c_h}$$

This expression provides a clear economic interpretation of the inventory decision:

- If the shortage cost c_s is significantly higher than the holding cost c_h , the critical fractile approaches 1, implying that the firm should stock more inventory to avoid costly stockouts.
- Conversely, if the holding cost c_h dominates, the critical fractile decreases, leading to a lower optimal inventory level to minimize excess stock.

Thus, the optimal policy is not driven solely by average demand, but by the trade-off between the costs of overstocking and understocking. This highlights the importance of accurately estimating cost parameters in practical inventory management.

Numerical Illustration

To demonstrate the practical implementation of the model, consider the following parameter values:

$$\mu = 20000, \quad \sigma = 5000, \quad c_h = 2, \quad c_s = 8$$

The critical fractile is given by:

$$\frac{c_s}{c_s + c_h} = \frac{8}{10} = 0.8$$

Using standard normal distribution tables, we find:

$$z^* \approx 0.84 \quad \text{such that} \quad \Phi(z^*) = 0.8$$

Thus, the optimal order quantity is:

$$Q^* = \mu + z^* \sigma = 20000 + (0.84)(5000) = 24200$$

This result indicates that the optimal inventory level exceeds the mean demand due to the higher penalty associated with stockouts. The model therefore recommends maintaining a buffer stock to hedge against demand uncertainty.

Visual Representation

The descriptive charts highlight significant demand spikes during holiday weeks and showcase the correlation coefficients between external variables (like markdowns) and weekly sales.

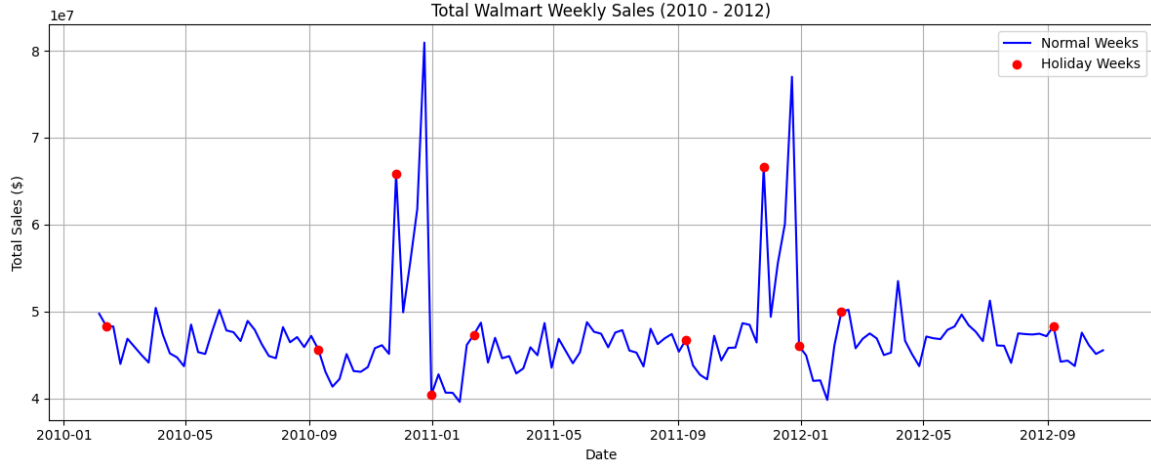


Figure 1: Total Walmart Weekly Sales (2010–2012). Red markers indicate holiday weeks where demand spikes are observed.

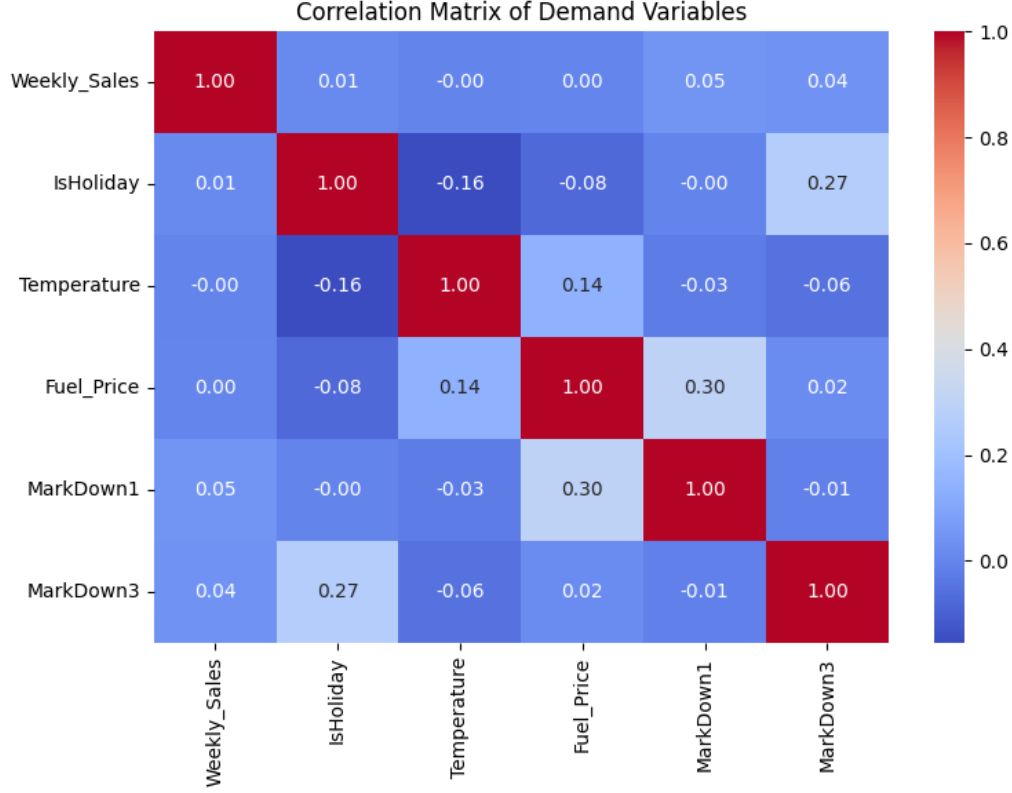


Figure 2: Correlation matrix of demand-related variables. Weak correlations suggest demand is influenced by multiple interacting factors.

The time-series plot reveals strong demand spikes during holiday periods, indicating significant seasonality. The correlation matrix shows that no single external variable dominates demand, reinforcing the need for stochastic modeling.

5 Conclusion

Interpretations and Conclusions

The statistical optimization approach provides a rigorous, data-driven methodology for retail inventory management. By moving away from deterministic averages and modeling demand as a stochastic random variable, we successfully minimized the expected financial exposure of the department. The convex nature of the problem ensures that the solution found is mathematically sound and globally optimal.

Limitations of the Study

First, assuming demand follows a perfect Normal distribution is a simplification; retail sales data often exhibits skewness and fat tails. Second, removing negative sales (returns) artificially smooths the operational reality. Finally, the model assumes holding and shortage costs are static linear functions, whereas in practice they may be nonlinear and time-varying.

Future Directions

Future work should integrate advanced machine learning techniques, such as Long Short-Term Memory (LSTM) networks, to dynamically forecast μ and σ . Additionally, extending the model to multi-item constraints could transform it into a more realistic quadratic or quadratically constrained quadratic program (QCQP).